CRD-Net: Causal Reliability Distillation for RGB-Thermal Semantic Segmentation

1. Objective

Train a single end-to-end RGB-Thermal semantic segmentation model, **CRD-Net**, and measure whether causal, class-conditional reliability-guided fusion improves validation mIoU over a standard RGB-T fusion baseline.

The experiment uses only RGB and thermal images at inference time. Ground-truth segmentation masks are used only during training for supervision.

2. Research Idea

Existing reliability-aware RGB-T methods usually estimate modality uncertainty or reliability from feature uncertainty, reconstruction error, or artificial degradation.

CRD-Net introduces **causal reliability distillation**:

- During training, compare the fused prediction with predictions obtained by removing or masking each modality.
- Measure how much RGB and thermal features causally affect the final segmentation decision.
- Use this causal contribution as supervision for a reliability estimator.
- At inference, predict reliability directly from RGB and thermal features.

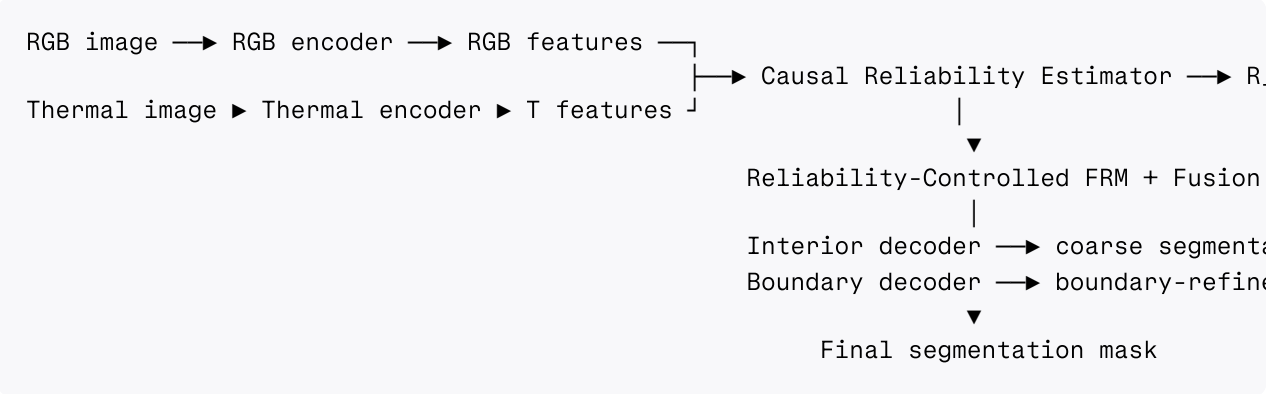
The key novelty is that reliability is learned from **counterfactual prediction changes**, not only from image degradation or uncertainty.

3. Proposed Contributions

1. **Causal Reliability Distillation (CRD)**
   Reliability is supervised by the change in fused segmentation output when a modality is removed.
2. **Class-Conditional Reliability**
   Instead of one reliability value per pixel, the model predicts separate RGB and thermal reliability scores for each semantic class.
3. **Reliability-Controlled Fusion Module (FRM)**
   RGB and thermal features are fused using predicted reliability weights.
4. **Boundary-Decoupled Refinement**
   A separate boundary-aware refinement branch improves segmentation near object edges, where mIoU is commonly lost.

5. **Single-Experiment Validation**
   The full model is trained once and evaluated on the validation set to verify mIoU improvement.

4. Architecture



4.1 Encoders

- RGB encoder extracts texture, color, and scene-context features.
- Thermal encoder extracts temperature-based object cues.
- Both encoders should use the same backbone family for fair feature alignment.

Recommended lightweight backbone:

- MiT-B0 or MiT-B1 for transformer-based design
- ResNet-18 for a faster and simpler baseline

4.2 Reliability Estimator

The reliability estimator receives RGB features, thermal features, and initial cross-modal interaction features.

It outputs:

$$R_{RGB} \in \mathbb{R}^{H \times W \times C}$$

$$R_T \in \mathbb{R}^{H \times W \times C}$$

where:

- H, W are feature-map height and width
- C is the number of semantic classes
- R_{RGB} and R_T are class-wise reliability maps

A sigmoid activation ensures reliability values remain between 0 and 1.

4.3 Reliability-Controlled FRM

For each pixel i and class c:

$$F_{fused}(i,c) = R_{RGB}(i,c)F_{RGB}(i,c) + R_T(i,c)F_T(i,c)$$

Before final fusion, normalize the two reliability values:

$$\hat{R}_{RGB}(i,c) = \frac{R_{RGB}(i,c)}{R_{RGB}(i,c) + R_T(i,c) + \epsilon}$$

$$\hat{R}_T(i,c) = \frac{R_T(i,c)}{R_{RGB}(i,c) + R_T(i,c) + \epsilon}$$

This prevents both reliabilities from becoming simultaneously high or low.

4.4 Boundary Refinement

A lightweight boundary branch uses gradient information from RGB and thermal images:

$$B = \text{Sobel}(I_{RGB}) + \lambda \text{Sobel}(I_T)$$

The boundary branch applies stronger cross-modal interaction near edges and refines the coarse segmentation output.

5. Training Strategy

5.1 Causal Reliability Supervision

For each training image, obtain:

- P_{RGB}: RGB-only prediction
- P_T: thermal-only prediction
- P_F: fused RGB-T prediction
- $P_{\text{no RGB}}$: fused prediction with RGB contribution removed
- $P_{\text{no T}}$: fused prediction with thermal contribution removed

Compute causal contribution using KL divergence:

$$C_{RGB}(i,c) = D_{KL}\left(P_F(i,c) \parallel P_{\text{no RGB}}(i,c)\right)$$

$$C_T(i,c) = D_{KL}\left(P_F(i,c) \parallel P_{\text{no T}}(i,c)\right)$$

The target reliability is:

$$R_m^*(i,c) = \sigma\left(\alpha C_m(i,c) - \beta U_m(i,c)\right)$$

where:

- $m \in \{RGB, T\}$

- U_m is the uncertainty or entropy of the modality-specific prediction
- α and β are weighting hyperparameters

The reliability head is trained to predict R_m^*.

5.2 Loss Functions

The total training loss is:

$$
[
\mathcal{L}_{total}
$$

\mathcal{L}*{seg}
+
\lambda_1\mathcal{L}*{rel}
+
\lambda_2\mathcal{L}*{bd}
+
\lambda_3\mathcal{L}*{cons}
+
\lambda_4\mathcal{L}_{cal}
]

Loss	Purpose
$\mathcal{L}_{seg}$	Cross-entropy + Dice/IoU loss for segmentation
$\mathcal{L}_{rel}$	Supervises predicted reliability using causal contribution
$\mathcal{L}_{bd}$	Boundary-aware loss for edge refinement
$\mathcal{L}_{cons}$	Ensures fusion is not worse than the best unimodal branch
$\mathcal{L}_{cal}$	Penalizes overconfident reliability estimates

The consistency loss is:

$$
[
\mathcal{L}_{cons}
$$

\max
\left(
0,
\mathcal{L}_{fused}

\min
\left(
\mathcal{L}_{RGB},
\mathcal{L}_T

\right)
\right)
]

6. Experimental Setup

6.1 Dataset

Use **MFNet** as the primary dataset.

- RGB-T image pairs: 1,569
- Semantic classes: 9
- Standard evaluation metric: mIoU
- Validation set: official MFNet test/validation split

6.2 Input and Training Configuration

Setting	Value
Input resolution	480 × 640
Batch size	2–4
Epochs	100–150
Optimizer	AdamW
Initial learning rate	6e-5 to 1e-4
LR scheduler	Poly decay
Weight decay	0.01
Backbone	MiT-B0/B1 or ResNet-18
GPU	Single GPU, preferably ≥ 12 GB VRAM
Framework	PyTorch

6.3 Data Augmentation

Use only stable augmentations:

- Random horizontal flip
- Random scale
- Random crop
- Color jitter for RGB only
- Gaussian noise for thermal only, applied lightly

Avoid aggressive augmentation in the first experiment because it may make reliability supervision noisy.

7. Evaluation Protocol

Evaluate the final trained model on the validation set using:

- Overall mIoU
- Per-class IoU
- Pixel accuracy
- Frequency-weighted IoU, if required
- Boundary F-score, optional
- Inference time and parameter count, optional

The main success metric is:

[
\Delta \text{mIoU} =
\text{mIoU}_{\text{CRD-Net}}

\text{mIoU}_{\text{Baseline}}
]

The experiment is successful if:

$$\Delta mIoU > 0$$

A practical target is an improvement of approximately **+1.0 to +2.5 mIoU** over the baseline.

8. Baseline

Use a standard RGB-T encoder-decoder segmentation network with:

- RGB encoder
- Thermal encoder
- Concatenation or attention-based fusion
- Segmentation decoder
- CE + Dice/IoU loss

The baseline does **not** include:

- Causal reliability estimator
- Class-conditional reliability
- Reliability-controlled FRM
- Boundary-decoupled refinement

9. Implementation Plan

Phase 1: Data Pipeline

1. Load RGB and thermal images.
2. Resize both to 480 × 640.
3. Normalize RGB and thermal images separately.
4. Apply identical geometric augmentation to both modalities and masks.
5. Return RGB, thermal, mask, and file metadata.

Phase 2: Baseline Model

1. Implement RGB and thermal encoders.
2. Add simple fusion: concatenation followed by convolution, or lightweight cross-attention.
3. Add segmentation decoder.
4. Train with CE + Dice/IoU loss.
5. Record baseline validation mIoU.

Phase 3: CRD-Net

1. Add modality-specific prediction heads.
2. Add causal reliability estimator.
3. Add class-conditional reliability maps.
4. Replace simple fusion with reliability-controlled FRM.
5. Add boundary refinement branch.
6. Add all losses.
7. Train the full model once.

Phase 4: Evaluation

1. Load the best validation checkpoint.
2. Run inference on the validation set.
3. Compute mIoU and per-class IoU.
4. Compare against baseline.
5. Save predictions, reliability maps, loss curves, and logs.

10. Expected Results

Model	Expected Outcome
Baseline RGB-T fusion	Reference mIoU
CRD-Net full model	Higher mIoU due to better modality weighting and boundary refinement
RGB-only model	Lower mIoU in low-light or texture-poor scenes
Thermal-only model	Lower mIoU for classes requiring color/texture cues

Expected qualitative improvements:

- Better person and vehicle segmentation using thermal cues
- Better road, sky, vegetation, and background segmentation using RGB cues
- Cleaner object boundaries
- More stable predictions when one modality is degraded

11. Ablation Plan for Later

After the single full experiment succeeds, perform these ablations one at a time:

Ablation	Removed Component	Purpose
A1	Causal reliability distillation	Tests whether causal supervision matters
A2	Class-conditional reliability	Tests whether class-wise reliability matters
A3	Boundary refinement	Tests whether boundary modeling improves IoU
A4	Consistency loss	Tests whether fusion stability improves performance
A5	Reliability-controlled FRM	Tests the importance of reliability-based fusion

12. Success Criteria

The experiment is considered successful if CRD-Net achieves:

1. Higher validation mIoU than the baseline.
2. Improvement in at least 3–4 per-class IoU scores.
3. Better performance under at least one degraded condition, such as low light, RGB blur, or thermal noise.
4. Reliability maps that visually correspond to modality usefulness.
5. No major training instability or overfitting.

13. Paper Claim

A safe novelty claim is:

> “We propose CRD-Net, a class-conditional causal reliability distillation framework for RGB-Thermal semantic segmentation. Unlike conventional uncertainty- or degradation-based reliability estimation, CRD-Net supervises modality reliability from counterfactual changes in fused segmentation predictions. The learned reliability controls a fusion module and a boundary-decoupled refinement decoder, improving segmentation accuracy without requiring ground-truth masks at inference time.”

14. Deliverables

- Trained CRD-Net checkpoint
- Baseline checkpoint
- Validation mIoU report
- Per-class IoU table
- Training and validation loss curves
- Reliability heatmap visualizations
- Segmentation prediction visualizations
- Boundary-refinement comparison images
- Experiment log with hyperparameters and random seed